

- 5 (a) The mean value of an alternating current is zero.  
Explain

- (i) Why an alternating current gives rise to a heating effect in a resistor,

.....  
 .....  
 ..... [2]

- (ii) by reference to heating effect, what is meant by the root-mean-square (r.m.s.) value of an alternating current.

.....  
 .....  
 ..... [1]

- (b) The variation with time  $t$  of the output  $V$  of an alternating voltage supply of frequency 50 Hz is shown in Fig. 5.1.

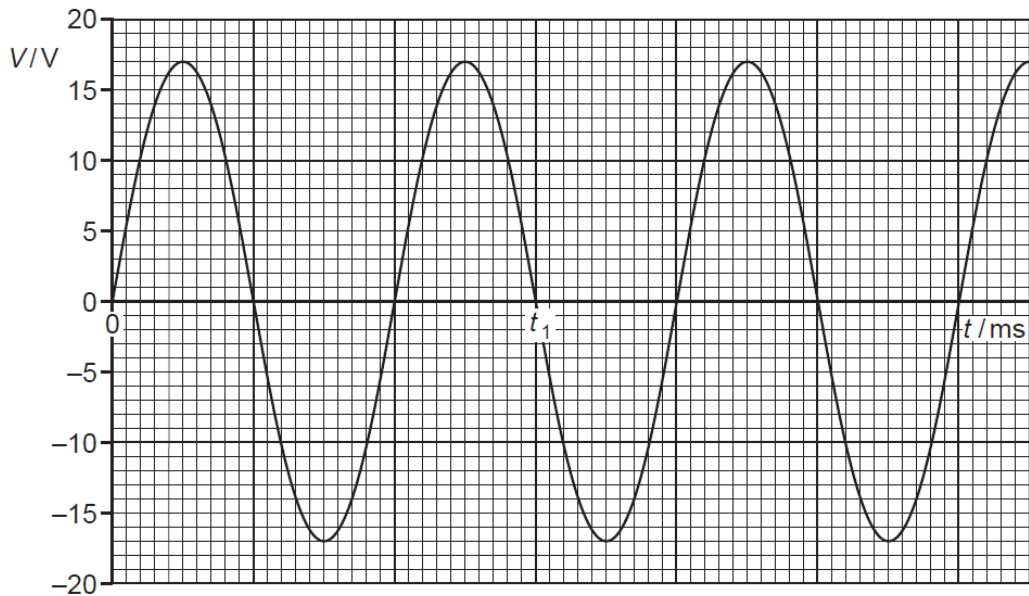


Fig. 5.1

Use Fig. 5.1 to determine

- (i) the time  $t_1$ ,

$t_1 = \dots\dots\dots$  s [2]

- (ii) the root-mean-square voltage  $V_{rms}$

$$V_{rms} = \dots\dots\dots \text{ V} \quad [2]$$

- (iii) The alternating supply is connected in series with a diode and a load resistor of resistance  $2.4 \, \Omega$ . On Fig. 5.2, sketch the variation with time  $t$  of the power  $P$  dissipated in the load resistor for time  $t = 0$  to  $t = 40 \text{ ms}$ . Assume that  $P = 0$  when  $t = 0$ . [3]

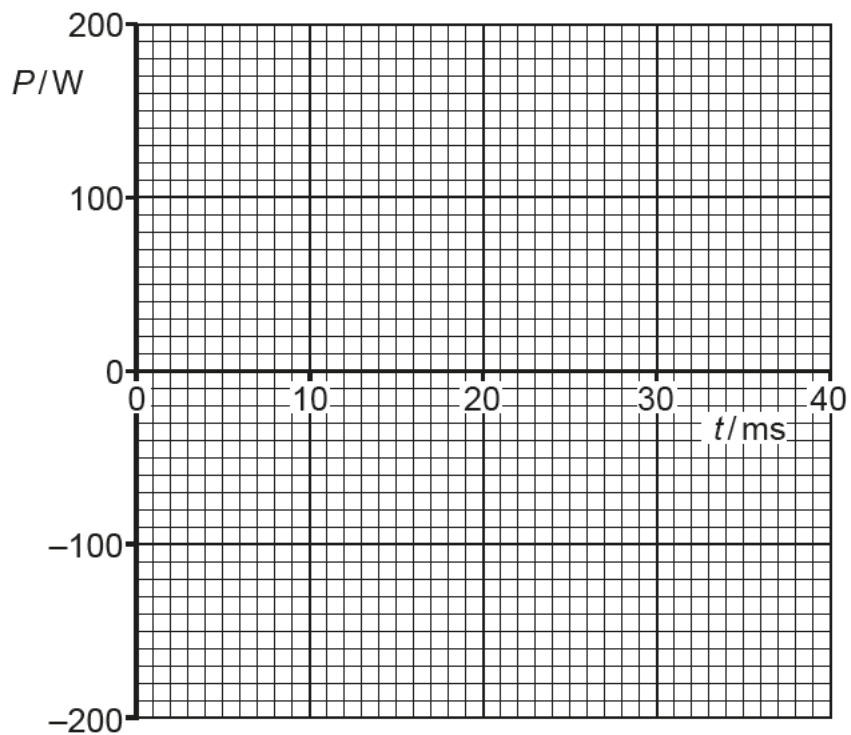


Fig. 5.2

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[ (c) Electromagnetic radiation of frequency  $f$  is incident on a metal surface. The variation